

Pseudocode

Algorithm: Baseline interval-masked top-k Pearson correlation

Input:

$X[F][S]$ // float values
 $start[F], end[F]$ // valid interval of feature i : $[start[i], end[i])$
 k // number of neighbors to return per feature

Output:

For each i : $TopK[i] = k$ indices j ($j \neq i$) with largest $|corr(i,j)|$

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1: for i in 0..F-1 do
2:   si ← start[i]; ei ← end[i]
3:    $\mu[i] \leftarrow \text{mean}(X[i][si:ei])$ 
4:    $\sigma[i] \leftarrow \text{std\_sample}(X[i][si:ei])$  // sample std (uses n-1 in variance)
5: end for

6: for i in 0..F-1 do
7:   for j in 0..F-1 do
8:     if j == i then
9:        $corr[i][j] \leftarrow 0$ 
10:      continue
11:    end if

12:    L ← max(start[i], start[j])
13:    R ← min(end[i], end[j])
14:    n ← R - L
15:    if n < 2 then
16:       $corr[i][j] \leftarrow 0$ 
17:      continue
18:    end if

19:    acc ← 0
20:    for t in L..R-1 do
21:       $z_i \leftarrow (X[i][t] - \mu[i]) / \sigma[i]$ 
22:       $z_j \leftarrow (X[j][t] - \mu[j]) / \sigma[j]$ 
23:      acc ← acc +  $z_i * z_j$ 
24:    end for

25:     $corr[i][j] \leftarrow acc / (n - 1)$  // Pearson correlation over overlap
26:  end for
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27: end for

28: for i in 0..F-1 do
29:     TopK[i] ← indices of k largest values of |corr[i][j]| over all j !
    = i
30: end for

31: return TopK
```

Notes:

- mean/std are computed per feature over its own valid interval.
- corr(i,j) is computed over the overlap interval [L,R).